1.Create a simple text file called suspicious_code.txt that contains a harmless script resembling a Trojan horse (for example, a script that pretends to be a game but actually displays a fake 'You have been hacked!' message when run).

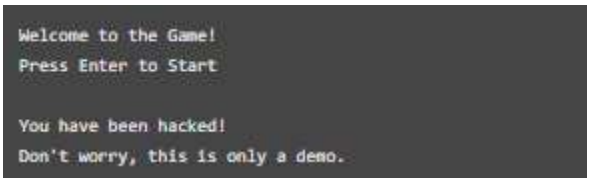
```
print("Welcome to the Game!")

print("Press Enter to Start")


input()


print("You have been hacked!")

print("Don't worry, this is only a demo.")
```



**2.Write a Python script named worm_simulator.py that copies itself into a new file with a different name (e.g., worm_copy1.py) in the same directory every time it runs.

Constraint: Limit the worm to make only 3 copies to avoid cluttering your folder.**

```
import os

import shutil


source = os.path.abspath(__file__)

folder = os.path.dirname(source)


for i in range(1, 4):

   new_file = os.path.join(
```

```
        folder, f"worm_copy{i}.py"
    )

    if not os.path.exists(new_file):
        shutil.copyfile(source, new_file)
        print("Created:", os.path.basename(new_file))
        break
else:
    print("Maximum 3 copies already exist.")
```



3.Simulate a virus infection by creating a batch file (virus_sim.bat) that appends a line of text to every .txt file in a given folder, mimicking how a virus might spread.

```
@echo off
set "folder=%~dp0test_files"

if not exist "%folder%" mkdir "%folder%"

if not exist "%folder%\file1.txt" (
    echo This is test file 1 > "%folder%\file1.txt"
)

if not exist "%folder%\file2.txt" (
```

```
    echo This is test file 2 > "%folder%\file2.txt"
)

for %%f in ("%folder%\*.txt") do (
    echo Virus Simulation Test >> "%%~f"
)

echo Simulation completed.
Pause
```



4.Research and list 3 real-world examples each of Trojans, viruses, and worms that have affected popular platforms or apps (such as WhatsApp, Windows, or Android). For each, write one line about how it spread or what damage it caused.

1. Trojan Horse Examples

Trojan	**How it spread or caused damage**
Zeus	Stole banking information from infected Windows computers.
Emotet	Spread through malicious email attachments and downloaded more malware.
Triada	Infected Android devices and could steal information or install unwanted apps.

2. Virus Examples

Virus	**How it spread or caused damage**
CIH (Chernobyl)	Infected Windows executable files and could damage system data.
Sality	Infected Windows programs and weakened computer security.

Virut	Infected executable files and connected infected computers to malicious networks.

3. Worm Examples

Worm	**How it spread or caused damage**
WannaCry	Spread through vulnerable Windows systems and encrypted files.
Conficker	Spread through Windows network vulnerabilities and infected many computers.
Morris Worm	Spread across Unix computers and caused network disruption.

**5.Given the following code snippet that is supposed to detect a worm by checking for duplicate files in a folder, find and fix the bug so it correctly identifies all worm copies:

import os
files = os.listdir('.')
for f in files:
  if 'worm_copy' in f:
  print('Worm found:', f)

Hint: The code misses the original worm file and might not catch all copies if the naming changes. Adjust the condition to be more flexible.**

```
import os
import hashlib

def get_hash(filename):
  with open(filename, "rb") as file:
    return hashlib.sha256(file.read()).hexdigest()

original = "worm_simulator.py"

if not os.path.isfile(original):
```

```
    print("Original worm file not found.")
else:
    original_hash = get_hash(original)

    for f in os.listdir("."):
        if not os.path.isfile(f):
            continue

        if f == original:
            print("Original found:", f)

        elif f.endswith(".py"):
            if get_hash(f) == original_hash:
                print("Worm copy found:", f)
```

```
Original found: worm_simulator.py
Worm copy found: worm_copy1.py
Worm copy found: worm_copy2.py
Worm copy found: worm_copy3.py
```